PROBLEM 13

Conservative Force and Potential IV

Original statement · Problems.pdf, p. 9

Quadratures & Trajectory

A particle of mass m moves along the Ox axis under the potential:

$$V(x) = mgx\left[\left(\frac{x}{a}\right)^2 - 3\right]$$

Compute the evolution of the particle as a function of time, when it is released from $x_0 = a$ with an initial velocity $v_0 = \sqrt{8ga}$ directed along the positive Ox axis.

(Tip: $\int \frac{dx}{(x+1)\sqrt{2-x}} = -\frac{2}{\sqrt{3}}\operatorname{arctanh}\left(\sqrt{\frac{2-x}{3}}\right)$*).*

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m on the Ox axis with the potential $V(x) = mgx\left[(x/a)^2 - 3\right]$, released from $x_0 = a$ with $v_0 = +\sqrt{8ga}$.
- **Constraints:** motion along the axis (smooth guide); $\mathrm{CDOF} = 1$, coordinate x. The dimensionless coordinate $y = x/a$ simplifies the algebra.
- **Parameters:** m [kg], g [m/s^2], a [m]. The statement gives the integral $\displaystyle\int \frac{dx}{(x+1)\sqrt{2-x}} = -\frac{2}{\sqrt{3}}\operatorname{arctanh}\left(\frac{\sqrt{2-x}}{\sqrt{3}}\right)$.

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ along the axis.

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i} = ay\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i} = a\dot{y}\,\mathbf{i}, \qquad \mathbf{a} = a\ddot{y}\,\mathbf{i}$$

3.2 Forces

- **Applied force:** $F = -V'(x)$ (conservative); **constraint forces** perpendicular to Ox; **inertia forces:** none.

Pedagogical Justification: the equilibrium points and their stability give the energy landscape. In terms of y, $V = mga\left(y^3 - 3y\right)$:

$$\frac{dV}{dy} = 3mga\left(y^2 - 1\right) = 0 \iff y = \pm 1, \qquad \frac{d^2V}{dy^2} = 6mga\,y$$

- $y = +1$ ($x = a$): $V = -2mga$, $V'' = +6mga > 0$: local minimum (stable). $y = -1$ ($x = -a$): $V = +2mga$, $V'' = -6mga < 0$: local maximum (unstable).
- The force follows from the chain rule, $F = -\dfrac{dV}{dx} = -\dfrac{1}{a}\dfrac{dV}{dy} = -3mg\left(y^2 - 1\right)$ (units of force: $[mg] = \mathrm{N}$).

3.3 Equations of Motion

- **Newton's second law:** $ma\ddot{y} = -3mg\left(y^2 - 1\right)$, i.e. $\ddot{y} = -\dfrac{3g}{a}\left(y^2 - 1\right)$ (one unknown, one equation). The energy is a first integral, as in Problem 12.
- **Total energy** from the initial state $y = 1$, $\dot{x} = \sqrt{8ga}$:

$$E = \tfrac{1}{2}m(8ga) + mga(1 - 3) = 4mga - 2mga = 2mga$$

It equals the height of the barrier, $V(-a) = +2mga$: the initial energy is exactly the separatrix energy.

3.4 Integration and Answers

Quadrature

> **Pedagogical Justification:** the energy equation gives a separable first-order equation. With $x = ay$:

$$\tfrac{1}{2}ma^2\dot{y}^2 = E - V = 2mga - mga\left(y^3 - 3y\right) = mga\left(2 + 3y - y^3\right) \implies \dot{y}^2 = \frac{2g}{a}\left(2 + 3y - y^3\right)$$

- **Factorisation.** $y = -1$ is a root of $y^3 - 3y - 2$ ($-1 + 3 - 2 = 0$). Dividing by $(y + 1)$ (check: $(y + 1)(y^2 - y - 2) = y^3 - y^2 - 2y + y^2 - y - 2 = y^3 - 3y - 2$): $y^3 - 3y - 2 = (y + 1)\left(y^2 - y - 2\right) = (y + 1)(y + 1)(y - 2)$. Therefore

$$2 + 3y - y^3 = -(y + 1)^2(y - 2) = (y + 1)^2(2 - y), \qquad \dot{y}^2 = \frac{2g}{a}(y + 1)^2(2 - y)$$

Stage 1: from $y = 1$ to the turning point $y = 2$ ($0 \le t \le t_1$)

> **Pedagogical Justification:** initially $\dot{y}(0) = +\sqrt{8g/a} > 0$ and for $1 \le y \le 2$ both $y + 1$ and $2 - y$ are non-negative, so the positive root is taken. Separating variables and applying Barrow's rule from $y(0) = 1$:

$$\int_1^y \frac{dy'}{(y' + 1)\sqrt{2 - y'}} = \sqrt{\frac{2g}{a}} \int_0^t dt' = \sqrt{\frac{2g}{a}}\, t$$

- With the integral of the statement, $\left[-\frac{2}{\sqrt{3}} \operatorname{arctanh} \sqrt{\frac{2 - y'}{3}}\right]_1^y$, and the lower limit $\sqrt{(2 - 1)/3} = 1/\sqrt{3}$:

$$-\frac{2}{\sqrt{3}}\operatorname{arctanh}\sqrt{\frac{2-y}{3}}+\frac{2}{\sqrt{3}}\operatorname{arctanh}\frac{1}{\sqrt{3}}=\sqrt{\frac{2g}{a}}\,t$$

Multiplying by $-\sqrt{3}/2$, with $\dfrac{\sqrt{3}}{2}\sqrt{\dfrac{2g}{a}}=\sqrt{\dfrac{3g}{2a}}$:

$$\boxed{\operatorname{arctanh}\sqrt{\frac{2-y}{3}}=-\sqrt{\frac{3g}{2a}}\,t+\operatorname{arctanh}\frac{1}{\sqrt{3}}},\qquad 0\le t\le t_1$$

- **Turning point.** The velocity vanishes when $2-y=0$, i.e. $x=2a$, where $\operatorname{arctanh}(0)=0$:

$$0=-\sqrt{\frac{3g}{2a}}\,t_1+\operatorname{arctanh}\frac{1}{\sqrt{3}}\implies t_1=\sqrt{\frac{2a}{3g}}\operatorname{arctanh}\frac{1}{\sqrt{3}}$$

Stage 2: return motion ($t>t_1$)

> **Pedagogical Justification:** at $x=2a$ the force is $F=-3mg\,(4-1)=-9mg<0$, so the particle turns back and $\dot{y}<0$. The negative root is taken and the integral restarts at $(y=2, t=t_1)$:

$$\int_2^{y}\frac{dy'}{(y'+1)\sqrt{2-y'}}=-\sqrt{\frac{2g}{a}}\int_{t_1}^{t}dt'=-\sqrt{\frac{2g}{a}}\,(t-t_1)$$

$$-\frac{2}{\sqrt{3}}\operatorname{arctanh}\sqrt{\frac{2-y}{3}}-0=-\sqrt{\frac{2g}{a}}\,(t-t_1)\implies\operatorname{arctanh}\sqrt{\frac{2-y}{3}}=\sqrt{\frac{3g}{2a}}\,(t-t_1)$$

- Since $\sqrt{3g/(2a)}\,t_1=\operatorname{arctanh}\left(1/\sqrt{3}\right)$:

$$\boxed{\operatorname{arctanh}\sqrt{\frac{2-y}{3}} = \sqrt{\frac{3g}{2a}}\,t - \operatorname{arctanh}\frac{1}{\sqrt{3}}}, \qquad t \geq t_1$$

Position as a function of time

Pedagogical Justification: inverting both branches ($\sqrt{(2-y)/3} = \tanh(\cdot)$, then squaring) gives one function, because $\tanh^2$ is even:

$$\boxed{x(t) = a\,y(t) = a\left[2 - 3\tanh^2\left(\sqrt{\frac{3g}{2a}}\,t - \operatorname{arctanh}\frac{1}{\sqrt{3}}\right)\right]} \qquad \text{for all } t \geq 0$$

At $t = 0$, $\tanh\left(-\operatorname{arctanh}(1/\sqrt{3})\right) = -1/\sqrt{3}$ and $x(0) = a\left(2 - 3\cdot\tfrac{1}{3}\right) = a$; at $t = t_1$, $x = 2a$; and as $t \to \infty$, $\tanh^2 \to 1$ and $x \to -a$.

Interpretation & Checks

- **Asymptotic behaviour ($t \to \infty$):** the right-hand side of the Stage 2 equation grows without bound, and $\operatorname{arctanh}(u) \to +\infty$ only as $u \to 1^-$. So $\sqrt{(2-y)/3} \to 1$, i.e. $2 - y \to 3$ and $y \to -1$: the particle approaches the unstable maximum $x = -a$ asymptotically in infinite time and never crosses it. This follows from $E = V(-a)$ (separatrix).
- **Return through the well:** at $x = a$ the speed is again $\sqrt{\frac{2}{m}\left[E - V(a)\right]} = \sqrt{\frac{2}{m}\left[2mga + 2mga\right]} = \sqrt{8ga}$.
- **Dimensional check:** $[\dot{y}^2] = [2g/a] = \mathrm{s}^{-2}$, the argument of arctanh is dimensionless, and $[t_1] = \sqrt{[a]/[g]} = \mathrm{s}$.

- **Check against the official key:** the key of `Problems.pdf` gives exactly the two branches above, $\operatorname{arctanh}\sqrt{(2-y)/3} = -\sqrt{3g/(2a)}\,t + \operatorname{arctanh}(1/\sqrt{3})$ for $0 < t < t_1$ and its mirror for $t > t_1$, and the initial speed $\sqrt{8ga}$ (which it calls the terminal velocity). Numerical integration of $\ddot{y} = -(3g/a)\left(y^2 - 1\right)$ with $g = a = 1$ reproduces the closed form to 10^{-8} ($y(0.3) = 1.759553$, turning point $y = 2$ at $t_1 = 0.537646$, $y(2) = -0.683994$).

Source: Problems.pdf